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The two-equal-disjoint path cover problem of Matching Composition Network ☆

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Abstract

Embedding of paths have attracted much attention in the parallel processing. Many-to-many communication is one of the most central issues in various interconnection networks. A graph G is globally two-equal-disjoint path coverable if for any two distinct pairs of vertices (u,v) and (w,x) of G, there exist two disjoint paths P and Q satisfied that (1) P (Q, respectively) joins u and v (w and x, respectively), (2) $|P|=|Q|$, and (3) $V(P\cup Q)=V(G)$. The *Matching Composition Network* (MCN) is a family of networks which two components are connected by a perfect matching. In this paper, we consider the globally two-equal-disjoint path cover property of MCN. Applying our result, the Crossed cube CQ_n, the Twisted cube TQ_n, and the Möbius cube MQ_n can all be proven to be globally two-equal-disjoint path coverable for $n \geqslant 5$.

Keywords: Interconnection networks; Matching Composition Networks; k disjoint path cover; Two-equal-disjoint path cover; Hamiltonian-connectivity

1. Introduction

For the graph definition and notation, we follow [1]. $G=(V,E)$ is a *graph* if V is a finite set and E is a subset of $\{(u,v) \mid (u,v)$ is an unordered pair of $V\}$. We say that V is the *vertex set* and E is the *edge set*. Two vertices u and v are *adjacent* if $(u,v)\in E$. A *path*, denoted by $\langle v_0,v_1,v_2,\ldots,v_k\rangle$, is a sequence of distinct vertices where v_i and v_{i+1} are adjacent for all i, $0\leqslant i\leqslant k-1$. The *length* of a path P, $|P|$, is the number of edges in P. We also write the path $\langle v_0,v_1,v_2,\ldots,v_k\rangle$ as $\langle v_0,P_1,v_i,v_{i+1},\ldots,v_j,P_2,v_t,\ldots,v_k\rangle$, where P_1 is the subpath $\langle v_0,v_1,\ldots,v_i\rangle$ and P_2 is the subpath $\langle v_j,v_{j+1},\ldots,v_t\rangle$. It is possible to write a path as $\langle v_0,v_1,P,v_1,v_2,\ldots,v_k\rangle$ if $|P|$ is 0. Sometimes, a path can also be represented by $\langle v_0,v_1,\ldots,v_i,e,v_{i+1},\ldots,v_n\rangle$ to emphasize that e is the edge (v_i,v_{i+1}). We use $d(u,v)$ to denote the distance between u and v, i.e., the length of a shortest path joining u and v. A path is a *Hamiltonian path* if its vertices are distinct and span V. A *cycle* is a path with at least three vertices such that the first vertex is the same as the last one. A cycle is called a *Hamiltonian cycle* if its vertices are distinct except for the first vertex and the last vertex and if they span V. A *Hamiltonian graph* is a graph with

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a Hamiltonian cycle. A graph G is *Hamiltonian connected* if there exists a Hamiltonian path joining any two distinct vertices. A graph G is k-fault Hamiltonian (k-fault Hamiltonian connected, respectively) if it remains Hamiltonian (Hamiltonian connected, respectively), after removing at most k vertices and/or edges [8–10].

Finding vertex-disjoint paths is one of the important issues of routing among vertices in various interconnection networks. Vertex-disjoint (abbreviated as disjoint) paths can be used to avoid communication congestion and provide parallel paths for an efficient data routing among vertices. Moreover, multiple disjoint paths can be more fault-tolerant of vertex or edge failures and greatly enhance the transmission reliability. Disjoint paths generally fall into three categories: one-to-one, one-to-many, and many-to-many. The one-to-one disjoint path is built with one source and one destination. The one-to-many disjoint paths like a tree structure, they contain one source and many distinct destination vertices. The many-to-many disjoint paths involve k, $k \geqslant 1$, disjoint paths with k pairs distinct source and destination vertices.

A *disjoint path cover* in a graph G is a set of disjoint paths containing all the vertices in G. For an embedding of linear arrays in a network, the cover implies every vertex can be participated in a pipeline computation. One-to-one disjoint path cover in recursive circulants [13] and one-to-many disjoint path cover in some hypercube-like interconnection networks [14] were studied. The many-to-many k-disjoint path cover is proposed by Park et al. in [15]. In this paper, we study a variation of many-to-many k-disjoint path cover (abbreviated as k-disjoint path cover) property as many-to-many k-equal-disjoint path cover (abbreviated as k-equal-disjoint path cover) that k disjoint paths have the same length. The k disjoint paths with equal length implies that the parallel processing of k pipelines is guaranteed accurately. Furthermore, a graph is called globally k-equal-disjoint path coverable if there exists a k-equal-disjoint path cover for any k distinct source-destination pairs.

In this paper, we study the two-equal-disjoint path cover property of a family of interconnection networks, called the Matching Composition Networks (MCN), which can be recursively constructed. MCN includes many well-known interconnection networks as special cases, such as the Hypercube Q_n, the Crossed cube CQ_n, the Twisted cube TQ_n, and the Möbius cube MQ_n. Basically, MCN and these mentioned cubes are all constructed from two graphs G_0 and G_1 with the same number of vertices, by adding a perfect matching between the vertices of G_0 and G_1. We shall call these two graphs G_0 and G_1 as the *components* of MCN.

The rest of this paper is organized as follows: The next section is preliminary. We give the formal definitions of the two-equal-disjoint path cover property and the Matching Composition Network. In Section 3, we discuss the main theorem. We then discuss the two-equal-disjoint path cover property of CQ_n, TQ_n, and MQ_n in Section 4. Finally, our conclusions are given in Section 5.

2. Preliminary

In this section, we will first give the definition of globally two-equal-disjoint path cover property of a graph G, and then we will give the relevant definitions in graph theory and the definition of the Matching Composition Networks.

Definition 1. A graph G is (u, v, w, x)-two-equal-disjoint path coverable if there are two disjoint paths P and Q with end vertices u, v and w, x, respectively, such that $|P| = |Q|$ and $V(P) \cup V(Q) = V(G)$.

Definition 2. A graph G is globally two-equal-disjoint path coverable if for any two distinct pairs of vertices (u, v) and (w, x), G is (u, v, w, x)-two-equal-disjoint path coverable.

For these two definitions, we have the following corollaries.

Corollary 1. *If a graph G is globally two-equal-disjoint path coverable then G is also Hamiltonian.*

Proof. Let P and Q be the (u, v, w, x)-two-equal-disjoint path cover of G such that P joins u to v and Q joins w to x where $(u, x), (v, w) \in E(G)$. Then $\langle u, P, v, w, Q, x, u\rangle$ is a Hamiltonian cycle of G. □

Corollary 2. *If a graph G is globally two-equal-disjoint path coverable then G is also Hamiltonian connected.*

Proof. Give any two arbitrary distinct vertices u and x of G. Let P and Q be the (u, v, w, x)-two-equal-disjoint path cover of G such that P joins u to v and Q joins w to x with $(v, w) \in E(G)$. Then $\langle u, P, v, w, Q, x\rangle$ is a Hamiltonian path of G. □

Now, we define the Matching Composition Network (MCN) as follows: Let G_0 and G_1 be two graphs

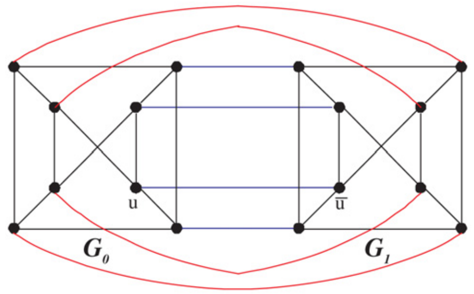

Fig. 1. An example of $G(G_0, G_1; L)$.

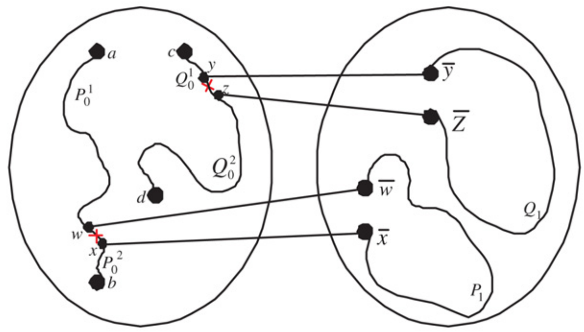

Fig. 2. a, b, c and d are in G_0.

with the same number of vertices. Let L be an arbitrary perfect matching between the vertices of G_0 and G_1, i.e., L is a set of edges connecting the vertices of G_0 and G_1 in a one-to-one fashion; the resulting composition graph is called a *Matching Composition Network* (MCN). Formally, we use the notation $G(G_0, G_1; L)$ to denote an MCN, which has vertices set $V(G(G_0, G_1; L)) = V(G_0) \cup V(G_1)$ and edge set $E(G(G_0, G_1; L)) = E(G_0) \cup E(G_1) \cup L$. Let u be a vertex in component G_i, $i = 0, 1$, of $G(G_0, G_1; L)$; and let $\bar{u}$ represent the neighbor of u in G_{1-i}. Please see Fig. 1 for an illustration.

It is not difficult to realize that $G(G_0, G_1; L)$ is Hamiltonian connected if both the two components are Hamiltonian connected. However, we expect that $G(G_0, G_1; L)$ is also globally two-equal-disjoint path coverable if both the two components are globally two-equal-disjoint path coverable. For example, the Crossed cube CQ_{n+1} is constructed from two copies of CQ_n by adding a perfect matching between the two and CQ_n is globally two-equal-disjoint path coverable for $n \geqslant 5$ [12].

3. Main result

We now ready to prove the globally two-equal-disjoint path cover property for Matching Composition Network.

Theorem 1. *Let G_0 and G_1 be two networks with the same number of vertices $N \geqslant 8$. Suppose that both of them are globally two-equal-disjoint path coverable. Then $G(G_0, G_1; L)$ is also globally two-equal-disjoint path coverable.*

Proof. Let (a, b) and (c, d) be two distinct source-destination pairs of $G(G_0, G_1; L)$. In the following, we establish two disjoint paths P, Q of length $N - 1$ with end vertices (a, b) and (c, d), respectively. By the relative positions of the four vertices, we divide the proof into four cases as follows.

Case 1: a, b, c and d are all in the same component G_i, $i = 0, 1$.

Without loss of generality, let a, b, c and d are all in G_0. By hypothesis, there are two disjoint paths $\langle a, P_0, b\rangle$ and $\langle c, Q_0, d\rangle$ in G_0, where $|P_0| = |Q_0| = \frac{N}{2} - 1$. Let (w, x) and (y, z) be two edge on P_0 and Q_0, respectively, and let $P_0 = \langle a, P_0^1, w, x, P_0^2, b\rangle$ and $Q_0 = \langle c, Q_0^1, y, z, Q_0^2, d\rangle$. By hypothesis again, we have two disjoint paths $\langle \overline{w}, P_1, \bar{x}\rangle$ and $\langle \bar{y}, Q_1, \bar{z}\rangle$ of length $\frac{N}{2} - 1$ in G_1. Let $P = \langle a, P_0^1, w, \overline{w}, P_1, \bar{x}, x, P_0^2, b\rangle$ and $Q = \langle c, Q_0^1, y, \bar{y}, Q_1, \bar{z}, z, Q_0^2, d\rangle$. Clearly, P and Q are two disjoint paths and $|P| = |Q| = N - 1$. (See Fig. 2.)

Case 2: a, b and c are in the same component G_i, $i = 0, 1$; d is in another component G_{1-i}.

Without loss of generality, let a, b, and c are in G_0 and d is in G_1. By the hypothesis $N \geqslant 8$, we can let w be a vertex in G_0 and $w \notin \{a, b, c, \bar{d}\}$. By hypothesis, there are two disjoint paths $\langle a, P_0, b\rangle$ and $\langle c, Q_0, w\rangle$ with $|P_0| = |Q_0| = \frac{N}{2} - 1$ in G_0. Let x be the neighbor of b on the path P_0 and let $P_0 = \langle a, P_0', x, b\rangle$. If $\bar{x} \neq d$ and $\bar{b} \neq d$, we can get two disjoint paths $\langle \bar{x}, P_1, \bar{b}\rangle$ and $\langle \overline{w}, Q_1, d\rangle$ with $|P_1| = |Q_1| = \frac{N}{2} - 1$ in G_1. (See Fig. 3.) Let $P = \langle a, P_0', x, \bar{x}, P_1, \bar{b}, b\rangle$ and $Q = \langle c, Q_0, w, \overline{w}, Q_1, d\rangle$. Clearly, P and Q are also two disjoint paths with $|P| = |Q| = N - 1$. Otherwise, the conditions $\bar{x} = d$ or $\bar{b} = d$ exist. Note that $\frac{N}{2} - 1 \geqslant 3$. Therefore, we can choose a and the neighbor of a on the path P_0 to replace b and x, respectively, and then rebuild another two-equal-disjoint path cover by the similar technique.

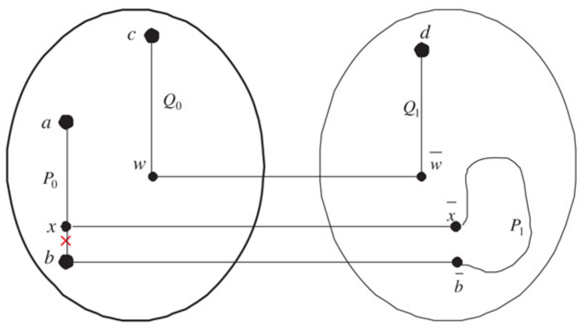

Fig. 3. a, b and c are in G_0; d is in G_1.

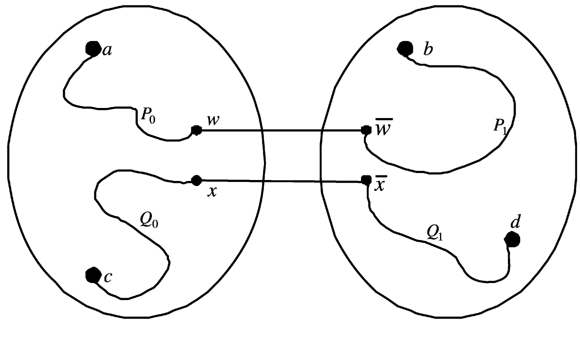

Fig. 4. a and c are in G_0; b and d are in G_1.

Case 3: a and b are both in the same component G_i, $i = 0, 1$; c and d are both in another component G_{1-i}.

Without loss of generality, let a and b be in G_0; and let c and d be in G_1. By hypothesis, there exists a Hamiltonian path P (Q, respectively) joining a (c, respectively) and b (d, respectively) in G_0 (G_1, respectively). Clearly, P and Q are two disjoint paths with $|P| = |Q| = N - 1$.

Case 4: a and c are both in the same component G_i, $i = 0, 1$; b and d are both in another component G_{1-i}.

Without loss of generality, let a and c be in G_0, and let b and d be in G_1. Let w and x be any two distinct vertices in G_0 except a and c with $\overline{w} \notin \{b, d\}$ and $\bar{x} \notin \{b, d\}$. By hypothesis, there are two disjoint paths $\langle a, P_0, w\rangle$ and $\langle c, Q_0, x\rangle$ with $|P_0| = |Q_0| = \frac{N}{2} - 1$ in G_0. Similarly, there are two disjoint paths $\langle \overline{w}, P_1, b\rangle$ and $\langle \bar{x}, Q_1, d\rangle$ with $|P_1| = |Q_1| = \frac{N}{2} - 1$ in G_1. Let $P = \langle a, P_0, w, \overline{w}, P_1, b\rangle$ and $Q = \langle c, Q_0, x, \bar{x}, Q_1, d\rangle$. Clearly, P and Q are also two disjoint paths with $|P| = |Q| = N - 1$. (See Fig. 4.) □

4. Applications

In this section, we demonstrate the usefulness of our proposed construction scheme for some well-known networks. For example, the Crossed cube CQ_n [3–5], the Twisted cube TQ_n [6,7], and the Möbius cube MQ_n [2] can all be proven to be globally two-equal-disjoint path coverable for $n \geqslant 5$.

The Hypercube is a popular topology for interconnection networks. The Crossed cube, the Twisted cube, and the Möbius cube are variations of the Hypercube. For each of these cubes, an n-dimensional cube can be constructed from two copies of $(n-1)$-dimensional subcubes by adding a perfect matching between the two subcubes. The main difference is that each of these cubes has various perfect matching between its subcubes. An n-dimensional cube has 2^n vertices, connectivity n, diagnosability n [11], and each vertex has the same degree n.

Hypercube is a well-known bipartite graph and there are no paths of odd length for any two end vertices belong to the same partite set. Note that $\frac{2^n}{2} - 1$ is odd number. That is, Q_n is not (u, v, w, x)-two-equal-disjoint path coverable for some vertices u, v, w, x. Therefore, Q_n is not globally two-equal-disjoint path coverable. In the following, we briefly state the recursive definitions of CQ_n, TQ_n, and MQ_n and prove they are all globally two-equal-disjoint path coverable for $n \geqslant 5$.

The vertices of these n-dimensional cubes are usually represented by the n-bit binary strings. A binary string u of length n will be written as $u = u_{n-1} u_{n-2}u_{n-3}\ldots u_0$, where $u_i \in \{0, 1\}$, $0 \leqslant i \leqslant n-1$. The following is the recursive definition of the n-dimensional Crossed cube CQ_n.

Definition 3. (See [3].) The Crossed cube CQ_1 is a complete graph with two vertices labeled by 0 and 1, respectively. For $n \geqslant 2$, an n-dimensional Crossed cube CQ_n consists of two $(n-1)$-dimensional sub-Crossed cubes, CQ^0_{n-1} and CQ^1_{n-1}, and a perfect matching between the vertices of CQ^0_{n-1} and CQ^1_{n-1} according to the following rule:

Let $V(CQ^0_{n-1}) = \{0u_{n-2}u_{n-3}\ldots u_0\colon\ u_i = 0 \text{ or } 1\}$ and $V(CQ^1_{n-1}) = \{1v_{n-2}v_{n-3}\ldots v_0\colon\ v_i = 0 \text{ or } 1\}$. The vertex $u = 0u_{n-2}u_{n-3}\ldots u_0 \in V(CQ^0_{n-1})$ and the vertex $v = 1v_{n-2}v_{n-3}\ldots v_0 \in V(CQ^1_{n-1})$ are adjacent in CQ_n if and only if

(1) $u_{n-2} = v_{n-2}$ if n is even, and
(2) $(u_{2i+1}u_{2i}, v_{2i+1}v_{2i}) \in \{(00, 00), (10, 10), (01, 11), (11, 01)\}$, for $0 \leqslant i < \lfloor\frac{n-1}{2}\rfloor$.

Hilbers et al. [7] defined the Twisted cubes using the parity function. Let $u = u_{n-1}u_{n-2}\dots u_0$, where $u_i \in \{0,1\}$ and $0 \leqslant i \leqslant n-1$, the parity function is defined as $P_i(u) = u_i \oplus u_{i-1} \oplus \cdots \oplus u_0$, where $\oplus$ is the *exclusive-or* operation.

Definition 4. (See [7].) The Twisted cube TQ_1 is a complete graph with two vertices, 0 and 1. Let n be an odd integer and $n \geqslant 3$. The vertices of an n-dimensional Twisted cube TQ_n are decomposed into four sets $S^{0,0}$, $S^{0,1}$, $S^{1,0}$ and $S^{1,1}$. The set $S^{i,j}$ consists of those vertices $u = u_{n-1}u_{n-2}\dots u_0$ with $u_{n-1} = i$ and $u_{n-2} = j$, where $(i,j) \in \{(0,0),(0,1),(1,0),(1,1)\}$. The induced subgraph of $S^{i,j}$ in TQ_n is isomorphic to TQ_{n-2}. Edges which connect these four $(n-2)$-dimensional subtwisted cubes can be described as follows: Any vertex $u_{n-1}u_{n-2}\dots u_0$ with $P_{n-3}(u) = 0$ is connected to $\bar{u}_{n-1}\bar{u}_{n-2}\dots u_0$ and $\bar{u}_{n-1}u_{n-2}\dots u_0$; and to $u_{n-1}\bar{u}_{n-2}\dots u_0$ and $\bar{u}_{n-1}u_{n-2}\dots u_0$, if $P_{n-3}(u) = 1$.

As stated in [7], for even integer n, the Twisted cube TQ_n can also be defined recursively in a similar way starting from TQ_2, where TQ_2 is isomorphic to Q_2. In order to see the recursive structure of the Twisted cube, we review the classical definition of the *Cartesian product*.

Definition 5. The *Cartesian product* of G and H, written $G \times H$, is the graph with vertex $V(G) \times V(H)$ specified by putting $\langle u,v\rangle$ adjacent to $\langle u',v'\rangle$ if and only if (1) $u = u'$ and $(v,v') \in E(H)$, or (2) $v = v'$ and $(u,u') \in E(G)$.

In Definition 4, let TQ^0_{n-1} (TQ^1_{n-1}, respectively) be the subgraph of TQ_n induced by $S^{0,0} \cup S^{1,0}$ ($S^{1,1} \cup S^{0,1}$, respectively). It follows directly from the definition that both TQ^0_{n-1} and TQ^1_{n-1} are isomorphic to the Cartesian product $TQ_{n-2} \times K_2$, where K_2 is the complete graph with two vertices. Then, TQ_n is constructed from TQ^0_{n-1} and TQ^1_{n-1} by joining them with a particular perfect matching.

We now present the definition of the Möbius cubes MQ_n [2]. There are two types of MQ_n, namely; $0-MQ_n$ and $1-MQ_n$.

Definition 6. (See [2].) $0-MQ_1$ and $1-MQ_1$ are both the complete graph on two vertices whose labels are 0 and 1. For $n \geqslant 2$, both $0-MQ_n$ and $1-MQ_n$ contain one 0-type sub-Möbius cube MQ^0_{n-1} and one 1-type sub-Möbius cube MQ^1_{n-1}. The first bit of every vertex of MQ^0_{n-1} is 0, and the first bit of every vertex of MQ^1_{n-1} is 1. For two vertices $u = 0u_{n-2}u_{n-3}\dots u_0 \in V(MQ^0_{n-1})$ and $v = 1v_{n-2}v_{n-3}\dots v_0 \in V(MQ^1_{n-1})$,

(1) u connects to v in $0-MQ_n$ if and only if $u_i = v_i$, for every i, $0 \leqslant i \leqslant n-2$,
(2) u connects to v in $1-MQ_n$ if and only if $u_i = \bar{v}_i$, for every i, $0 \leqslant i \leqslant n-2$.

Table 1
Some examples of not being (u,v,w,x)-two-equal-disjoint path coverable

Networks	(u,v)	(w,x)
CQ_3	(000, 001)	(010, 011)
CQ_4	(0000, 0001)	(0010, 1110)
$0-MQ_3$	(000, 001)	(010, 011)
$1-MQ_3$	(000, 001)	(010, 100)
$0-MQ_4$	(0000, 0001)	(0010, 1001)
$1-MQ_4$	(0000, 0011)	(0100, 1110)
TQ_3	(000, 001)	(010, 100)
TQ_4	(0000, 0001)	(0010, 1001)

Clearly, the numbers of vertices of CQ_n, TQ_n, and MQ_n are all not less than 8 for $n \geqslant 3$. However, we observe that CQ_3, CQ_4, TQ_3, TQ_4, MQ_3, and MQ_4 are not (u,v,w,x)-two-equal-disjoint path coverable for some source-destination pairs (u,v) and (w,x). Table 1 lists some examples.

Lemma 2. *CQ_5, TQ_5, and MQ_5 are all globally two-equal-disjoint path coverable.*

Proof. With long and detail discussion, we have completed theoretical proof for CQ_5, TQ_5, and MQ_5. Nevertheless, we do not present them in this paper for reducing complexity. However, we can also verify these small cases directly using computer by translating the original proof into a program. □

By Theorem 1, we can prove that CQ_n, TQ_n and MQ_n are all globally two-equal-disjoint path coverable for $n \geqslant 5$. Then, we list the following three theorems.

Theorem 3. *(See [12].) The Crossed cube CQ_n is globally two-equal-disjoint path coverable for $n \geqslant 5$.*

Theorem 4. *The Twisted cube TQ_n is globally two-equal-disjoint path coverable for $n \geqslant 5$.*

Theorem 5. *The Möbius cube MQ_n is globally two-equal-disjoint path coverable for $n \geqslant 5$.*

5. Conclusion

In this paper, we discussed the two-equal-disjoint path cover property of the Matching Composition Networks and propose a sufficient condition to verify if these networks are two-equal-disjoint path coverable. With the condition, we prove that the Crossed Cube CQ_n, the Twisted cube TQ_n, and the Möbius cube MQ_n are all globally two-equal-disjoint path coverable for $n \geqslant 5$. The globally two-equal-disjoint path cover problem is an extension of Hamiltonian-connectivity problem. We can see Hamiltonian-connectivity problem as globally one-path cover problem, and then we extended this property to globally two-equal-disjoint path coverable. This work may help to discuss the many-to-many disjoint path cover problem.

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